

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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**Flipped class room activity report on
Design of Digital Ammeter circuit using TinkerCAD
simulation tool**

**DEPARTMENT
OF
ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted by

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2025-2026

S.J.M VIDYAPEETHA ®

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE,
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1. Introduction

Measurement of electrical current is an essential requirement in electrical and electronic circuits. An ammeter is the instrument used to measure current flow, and traditional analog ammeters often suffer from limitations such as parallax error, mechanical wear, and lower accuracy. To overcome these drawbacks, digital ammeters are widely used because they provide precise readings, better reliability, and easy digital display. A digital ammeter works by converting the measured electrical quantity into a digital signal that can be processed and displayed accurately.

In this project, a Digital Ammeter is designed and simulated using **Arduino Uno** in the **TinkerCAD virtual designing tool**. The Arduino measures the voltage drop across a known resistor connected in series with the circuit and calculates the current using Ohm's Law. TinkerCAD offers a virtual environment to design, simulate, and test the circuit safely without physical components. This project helps in understanding current measurement principles, analog-to-digital conversion, and the practical use of microcontrollers in digital instrumentation systems.

2. Objectives

- To design and simulate a Digital Ammeter using Arduino in TinkerCAD
- To measure current flowing through a circuit accurately
- To understand the relationship between voltage, current, and resistance
- To display current values digitally using the Serial Monitor
- To gain hands-on experience with virtual circuit simulation

3. Software Requirements

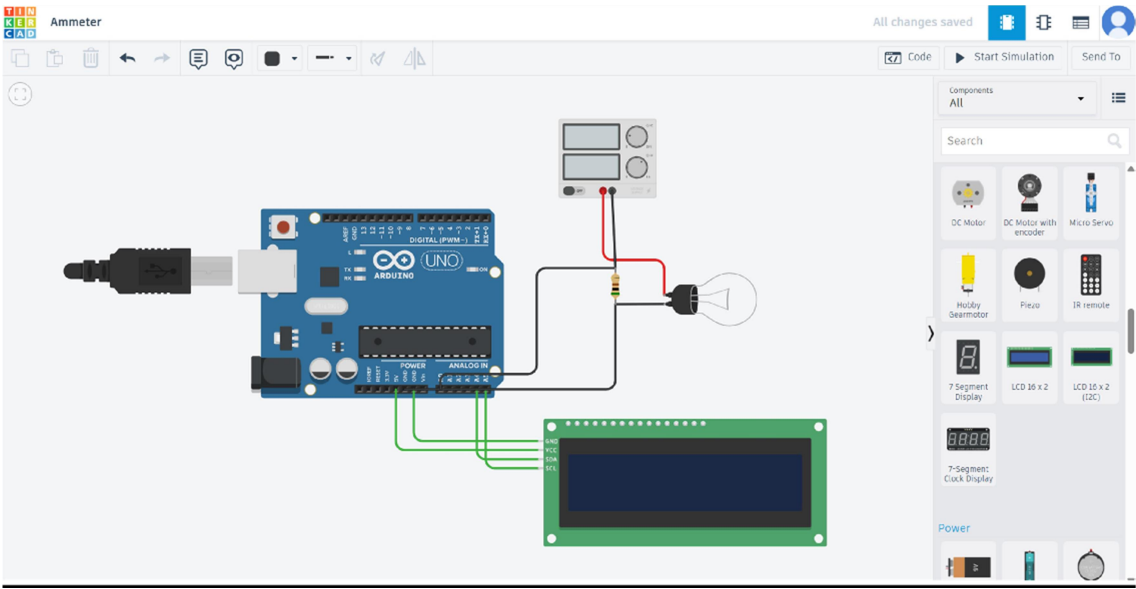
- TinkerCAD Circuits (Online Simulation Tool)
- Arduino Uno R3 (Virtual)
- Arduino IDE (built-in with TinkerCAD)
- Web Browser (Chrome / Edge / Firefox)
- Stable Internet Connection

4. Steps to Access TinkerCAD

1. Open a web browser and go to the TinkerCAD website
2. Sign in using a Google or Autodesk account
3. Click on **“Circuits”** from the dashboard
4. Select **“Create New Circuit”**
5. Drag and drop **Arduino Uno**, resistors, power supply, and connecting wires
6. Switch to **Code** → **Text mode** to write Arduino code
7. Start the simulation by clicking **“Start Simulation”**

Activity – Flipped Classroom

Circuit View

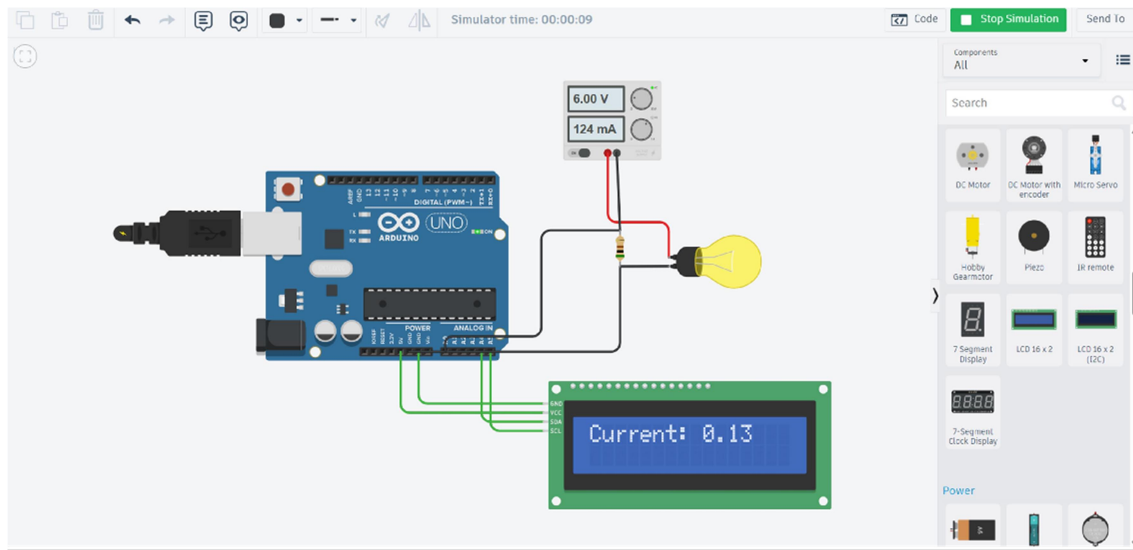


Components Required

Component List			Download CSV
Name	Quantity	Component	
L1	1	Light bulb	
U2	1	Arduino Uno R3	
R1	1	0.5 Ω Resistor	
P1	1	6V, 3.2A Power Supply	
U3	1	PCF8574-based, 39 (0x27) LCD 16 x 2 (I2C)	

Activity – Flipped Classroom

Circuit During Simulation



Required Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Initialize the LCD with I2C address and LCD size
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Define Pin
const int currentPin = A0;

// Constants
const float V_REF = 5.0;           // Reference Voltage of Arduino
const int ADC_RES = 1024;          // ADC Resolution
const float SHUNT_RESISTOR = 0.5;  // Shunt Resistance in ohms

void setup() {
    // Initialize the LCD
    lcd.init();
    lcd.backlight();

    // Print Initial Message
    lcd.setCursor(0, 0);
    lcd.print("Current: A");

    // Start Serial Communication
    Serial.begin(9600);
}

void loop() {
    // Read the Analog Value
    int currentRaw = analogRead(currentPin);

    // Convert Raw Value to Voltage
    float voltageAcrossShunt = (currentRaw * V_REF) / ADC_RES;

    // Calculate Current using Ohm's Law
    float current = abs(voltageAcrossShunt / SHUNT_RESISTOR);

    // Print Value to LCD
```

```
// Print Value to Serial Monitor
Serial.print("Current: ");
Serial.print(current, 2);
Serial.println(" A");

// Delay before next reading
delay(500);
}
```

5. Observations

1. The Arduino Uno successfully reads the analog voltage drop across the sensing resistor.
2. The analog input value changes proportionally with the variation in current.
3. The calculated current value increases when the load resistance is decreased.
4. The current value decreases when the load resistance is increased.
5. The Serial Monitor displays stable and continuous digital current readings.
6. Minor fluctuations in readings are observed due to ADC resolution limits.
7. The accuracy of the ammeter depends on the value and tolerance of the sensing resistor.
8. The system responds quickly to changes in circuit current during simulation.
9. The virtual simulation in TinkerCAD closely represents real-time circuit behavior.
10. The digital display eliminates parallax error, improving reading accuracy.

6. Result and Analysis

1. The Digital Ammeter circuit was successfully designed and simulated in TinkerCAD.
2. The Arduino Uno accurately measured the voltage drop across the sensing resistor.
3. The measured voltage was correctly converted into digital values using the ADC.
4. Current values were calculated accurately using Ohm's Law ($I = V / R$).
5. The displayed current readings varied proportionally with changes in load conditions.
6. The system produced stable and repeatable results during continuous operation.
7. Small variations in readings were observed due to ADC resolution and noise.
8. The simulation results closely matched theoretical current calculations.
9. The digital measurement method improved accuracy compared to analog ammeters.
10. The project demonstrated the effective use of Arduino for digital measurement applications.

7. Conclusion

1. The Digital Ammeter project was successfully designed and simulated using Arduino Uno in TinkerCAD.
2. The system accurately measured electrical current using voltage sensing and Ohm's Law.
3. The use of Arduino enabled precise analog-to-digital conversion of current values.
4. Digital display of current eliminated parallax and manual reading errors.
5. The project improved understanding of current measurement principles and ADC operation.
6. TinkerCAD provided a safe and cost-effective platform for circuit design and testing.
7. The obtained results closely matched theoretical calculations.
8. The project demonstrated reliable and stable performance during simulation.
9. The design can be easily modified for different current ranges and applications.
10. This project serves as a strong foundation for developing advanced digital instruments.